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List of Exhibits

The eight data sets analysed in Chapter 4 are supported by the following tables and figures. Each data set follows the same four-part structure: the table or graph, an explanation of the data, an analysis, and an interpretation.

Exhibit	Title	Data set
Table 1	Growth of UPI transactions, FY 2016-17 to FY 2025-26	Data Set 1
Figure 1	UPI transaction volume, FY 2016-17 to FY 2025-26	Data Set 1
Figure 2	UPI transaction value, FY 2016-17 to FY 2025-26	Data Set 1
Table 2	Average value of a single UPI transaction	Data Set 2
Figure 3	Average value of a single UPI transaction	Data Set 2
Table 3	UPI and total digital payments compared	Data Set 3
Figure 5	UPI's share of retail digital payment volume	Data Set 3
Table 4	UPI turnover relative to nominal GDP	Data Set 4
Figure 4	UPI turnover relative to nominal GDP	Data Set 4
Table 5	Digital payments around demonetisation	Data Set 5
Figure 6	Digital payments around demonetisation	Data Set 5
Table 6	UPI growth before, during and after COVID-19	Data Set 6
Figure 7	UPI volume before, during and after COVID-19	Data Set 6
Table 7	Indicators of financial inclusion and digitalisation	Data Set 7
Figure 8	Financial inclusion and payment digitalisation indices	Data Set 7
Figure 10	India's share of global real-time payment volume	Data Set 7
Table 8	Internet and telecom penetration, rural and urban	Data Set 8
Figure 9	The rural-urban digital divide	Data Set 8

1. Introduction

1.1 The meaning of digital payments

A payment means giving money to settle what you owe. For most of history this needed something physical to change hands. It could be a coin, a note or a cheque. A digital payment removes that object. The money moves as a message instead. The message goes between two bank accounts. The payer's balance goes down and the payee's balance goes up. No cash is touched at any point.

This difference matters more than it looks. Cash settles at once, but it moves slowly. It has to be withdrawn, carried, counted and then deposited again. A digital message travels at the speed of a phone network and settles in seconds. So it is not just easier to pay. The same money can also be used many more times in a year. That idea runs through this whole project.

1.2 The Unified Payments Interface

The Unified Payments Interface, or UPI, was launched in India in 2016. It is not an app. It is a standard, which means a common set of rules. These rules let any bank account be paid from any app. You use a short UPI address instead of an account number. So a user of one app can pay a shopkeeper who banks somewhere else and uses a different app. The payment is instant, free for both sides, and works at any time of day.

Three features explain why UPI worked when earlier systems did not. First, it is interoperable. This means the payer and the shopkeeper do not need the same app. Second, it never shuts down, unlike the older systems it replaced. Third, it is free for normal payments. Small shopkeepers pay no Merchant Discount Rate on payments up to ₹2,000. Instead, the Government pays the banks 0.15 per cent of the value of the payment [14].

The International Monetary Fund studied this design in June 2025. It said this was the main reason UPI spread so fast. Because any app works with any other, users are free to choose. Many people join through a brand they already trust and move to a better app later. Since users can leave at any time, every company has to keep improving. The result was a market with more than 200 apps, instead of one company taking everything [18].

1.3 The growth of digital transactions in India

The scale of the change is hard to show without figures. In FY 2016-17, its first full year, UPI handled about 2 crore transactions worth ₹0.07 lakh crore. In FY 2025-26 it handled over 24,162 crore transactions worth about ₹314 lakh crore. That is

about a twelve-thousand-fold rise in the number of payments in ten years. The value rose more than four thousand times [2].

This is not a small service used by a few people. UPI handled 81 per cent of all retail digital payments in India in FY 2024-25 [14]. The IMF calls it the largest retail fast payment system in the world by number of payments. It carries nearly 49 per cent of all real-time payments made anywhere in the world [3], [18]. By June 2026, 55.49 crore users had joined [16]. The number of banks on the platform rose from 44 to 703 by FY 2025-26 [2].

1.4 The role of the National Payments Corporation of India

None of this was left to the market. UPI was built and is run by the National Payments Corporation of India, or NPCI. NPCI is the umbrella body for retail payments in India. It was set up with the guidance of the Reserve Bank of India and the Indian Banks' Association. It is registered as a not-for-profit company. This matters, because NPCI does not have to earn a profit on the system itself. That is why payments can be carried at close to cost.

NPCI does five main jobs. It writes the technical rules. It runs the central switch that every payment passes through. It settles the money owed between member banks. It sets the limits on how much can be sent. It also runs the systems that watch for fraud while payments happen. Through its subsidiary, NPCI International Payments Limited, it has taken UPI abroad as well. UPI now works in eleven foreign countries. These include the United Arab Emirates, Singapore, Bhutan, Nepal, Sri Lanka, France, Mauritius and Qatar [17].

The fact that this is public infrastructure has drawn attention abroad. The Bank for International Settlements calls systems like UPI digital public infrastructure. It means open systems that anyone can join and that serve the whole country. The BIS says that Aadhaar and UPI together have made big gains in financial inclusion and in the efficiency of payments [20].

1.5 From a cash economy to a digital economy

India began this decade holding more cash than most countries, compared with the size of its economy. The Reserve Bank wants to bring that down. It named this as a goal in Payments Vision 2025 [26]. Cash is not free. It has to be printed, moved, guarded, sorted and destroyed. More importantly, a rupee sitting in a pocket or a cash box is doing no work at all.

So the move away from cash was pushed on purpose. It was not left to convenience. A survey done for the Department of Financial Services, released in February 2026, found that 90 per cent of users felt more confident about digital

payments after using UPI and RuPay. The same survey found a clear fall in how much cash they used and how often they visited ATMs [15]. This is what the national figures look like inside one household.

1.6 The meaning of the velocity of money

The velocity of money is the average number of times one rupee is used to buy goods and services in a given period. It is not a physical quality of money. It simply shows how quickly people spend the money they hold.

Irving Fisher explained this in his equation of exchange. It is usually written as:

$$M \times V = P \times T$$

Here M is the stock of money, V is velocity, P is the average price level and T is the number of transactions. The equation tells us something simple. If the stock of money stays the same and the number of transactions goes up, then velocity must have gone up too. There is also an income form of the equation, written as $V = (P \times Y) \div M$. Here $P \times Y$ is national income at current prices.

It is important to say which velocity we mean, and this project is careful about that. Income velocity compares national output with the money stock. It moves slowly. Transactions velocity compares the value of all payments with the money stock. It can move much faster. The reason is that transactions velocity counts every payment. That includes transfers which do not buy anything new. Payment systems affect transactions velocity directly, and income velocity only in an indirect way. Section 3.4 explains this in detail, because mixing up the two is the most common mistake on this topic.

1.7 Digital payments and economic activity

How could a payment system change real economic activity? There are four main ways. Each one comes up again in the research reviewed in Chapter 3.

1. Paying becomes cheaper. When a payment costs nothing and takes no time, people start making exchanges that were not worth the trouble before. This matters most for very small payments. Cash handled these badly, and cards never reached them at all.
2. People hold less idle money. A family that can pay straight from its bank account does not need much cash in hand. So the money held for spending falls compared with the amount actually spent. That is the same thing as a rise in velocity.
3. The informal economy becomes visible. A digital payment leaves a record. Records build up into a credit history. A credit history can turn a trader who

never had a loan into a borrower. The Reserve Bank has worked on this through the Unified Lending Interface [20].

4. Working capital is reused faster. A shopkeeper who is paid at once, instead of at the end of the day, can buy new stock sooner. The same rupee then pays for more sales in a year.

These are sensible explanations, not measured amounts. This project treats them that way throughout.

1.8 Financial inclusion and the formalisation of the economy

Financial inclusion means being able to use financial services that are useful and cheap. India built this in layers, and each layer needed the one before it. First came bank accounts for everyone under the Pradhan Mantri Jan Dhan Yojana. Then came a digital identity to prove who the account holder was. Last came a payment system those accounts could actually use. By 4 August 2025 the Jan Dhan scheme had reached over 55.98 crore people. More than 55 per cent of those accounts were held by women [7].

The result can be measured. The Reserve Bank's Financial Inclusion Index rose from 64.2 in March 2024 to 67.0 in March 2025. All three of its parts improved: access, usage and quality [7]. Formalisation comes from usage, not from access alone. An account that is only opened changes nothing. An account that is used creates a record of payments. Credit and insurance can then be built on that record.

1.9 Relevance in the post-demonetisation and post-pandemic economy

Two big shocks mark UPI's first ten years. In November 2016, ₹500 and ₹1,000 notes stopped being legal tender. They were worth ₹15.4 trillion. That was 86.9 per cent of the value of all notes then in circulation [23]. Monthly digital transactions rose 56 per cent between October 2016 and May 2017 [24]. UPI had launched only months earlier. So it was ready at the exact moment the country had to find something other than cash.

The second shock was COVID-19. It turned contactless payment into a health precaution rather than a convenience. It also pushed a whole generation of small shopkeepers into accepting a QR code. The two shocks worked in different ways, and Data Sets 5 and 6 study that difference. The first was a sudden force that people partly undid later. The second was a slower change that did not reverse.

1.10 Statement of the problem

Nobody doubts that UPI has grown. The figures come out every month and are not seriously questioned. The question I ask in this project is narrower and harder. Has that growth changed the speed at which money moves around in India? And can any effect on real economic activity be found in data that is public?

The gap between those two questions is what this report is about. A payment system can move a huge amount of money without changing national income at all. That happens if it only shifts payments that would have been made in cash anyway. To show that something more than a shift has happened, I need different evidence. I need to look at the size of payments, at who is paying, and at what shopkeepers say. That evidence is collected in Chapter 4.

2. Design of Study

2.1 Objectives of the study

This study has five objectives.

5. To understand what UPI is, how it is built, and how it grew in India between FY 2016-17 and FY 2025-26.
6. To study the link between UPI payments and the velocity of money, keeping transactions velocity and income velocity separate.
7. To study the effect of digital payments on economic activity. This includes spending habits, the sales of small shopkeepers, and the recording of payments.
8. To see how far UPI has helped financial inclusion, and how far it has brought informal activity into the recorded economy.
9. To study the problems and limits of digital payments in India. These include cyber fraud, dependence on networks, and uneven use across regions.

2.2 Research questions and hypotheses

These objectives are turned into three statements that can be tested. They are given here so the reader can judge, at the end of Chapter 4, whether the evidence supports them.

	Hypothesis	Evidence used
H1	The value of payments settled for every rupee of national output has risen sharply since FY 2016-17. In other words, transactions velocity has gone up.	Data Sets 1 and 4
H2	UPI's growth comes from new, small payments. It does not come only from large payments moving across from cash.	Data Sets 2 and 3
H3	The gains in inclusion and usage are not shared evenly between rural and urban India.	Data Sets 7 and 8

2.3 Data collection methodology

This study uses secondary data only. I did not carry out a survey, for a simple reason. The subject is national in size. It is already measured in full, every month,

by the bodies that run and regulate it. A survey done by a school student could not improve on those figures. Any sample small enough for me to collect would be far too small to stand for the whole country.

I chose sources in this order of preference:

- Operator and regulator data. Transaction figures from the National Payments Corporation of India. Also the Annual Report, Payment Systems Report, Digital Payments Index and Financial Inclusion Index of the Reserve Bank of India.
- Government releases. Press Information Bureau releases and factsheets, answers given in Parliament, and the Economic Survey.
- Official statistics. National income figures from the Ministry of Statistics and Programme Implementation. Telecom and internet data from the Telecom Regulatory Authority of India.
- International research. Work by the International Monetary Fund, the Bank for International Settlements and the World Bank. I used these mainly for comparison and for guidance on method.
- Academic research. Peer-reviewed papers and pre-prints. I used these where they show things about individual users that national figures cannot.

I avoided commercial and news websites as sources of figures. Every number quoted here comes from the body that produced it. The number in square brackets points to the entry in the Bibliography where that release can be found.

Two measures in this report were worked out by me, not taken from a source. The first is the average value of a UPI payment. I found it by dividing the yearly value by the yearly volume. The second is UPI turnover as a multiple of nominal GDP. I have written out both methods next to the tables so that anyone can check them.

2.4 Scope of the study

This study looks at the effect of digital payments on the Indian economy as a whole. Within that, it focuses on UPI. Other methods such as cards, wallets, IMPS, NEFT and RTGS appear only where a comparison is needed. The period runs from FY 2016-17, UPI's first full year, to FY 2025-26. That was the latest year with complete data when I wrote this. The main ideas studied are the velocity of money, consumption, and the recording of financial activity.

Some related questions are left out on purpose. This study does not look at the profits of payment companies. It does not look at the design of a central bank digital currency. It also does not look at the question of how few apps handle most UPI traffic. Each of those is a large topic on its own.

2.5 Limitations of the study

The limits below are stated openly. A study that hides them asks for more trust than its evidence deserves.

10. The study uses secondary data. Its conclusions can only be as reliable as the published figures behind them.
11. Correlation is not causation. UPI grew during a decade when smartphones, internet access, bank accounts and incomes all rose together. I can show that these things moved together. I cannot separate out UPI's own effect, and I do not claim to.
12. Velocity cannot be measured directly. It has to be worked out from the ratio of payments or income to a stock of money. Different choices give very different answers. The turnover ratio in Data Set 4 is an approximation, and I explain its limits where it appears.
13. Payment value is not output. A large share of UPI value is money sent between individuals. These transfers do not buy anything new, so they are not part of GDP. Any ratio of payment value to GDP has to be read with that in mind.
14. The national accounts were revised during this period. Nominal GDP is now published on a 2022-23 base. Earlier it used a 2011-12 base. So the two series do not join perfectly. I point this out wherever it affects a calculation.
15. The system changes quickly. Limits, fees and features were revised several times during the period. So the figures here are correct only up to the dates given.
16. Regional differences are hidden. National figures cover wide variation between states and districts. UPI data is not published at district level.

3. Review of Literature

This chapter sets out what is already known. It covers how digital payments developed, the theory of the velocity of money, and the research linking payments to economic activity. It also covers the policies behind UPI and the main criticisms made of it. These are the ideas and earlier findings I use when I do my own analysis in Chapter 4.

3.1 Meaning and evolution of digital payments

India did not reach UPI in one step. Each earlier system solved one problem and left another one behind.

Electronic transfers between banks came first, through RTGS and NEFT. These were reliable but were built for institutions. They worked in batches, or only during banking hours. The payer had to know the receiver's account number and branch code. They were meant for large payments. NPCI then brought in the Immediate Payment Service. This removed the limit of banking hours. For the first time, retail transfers could be made at any time of day.

Card networks solved a different problem, which was paying at a shop counter. But they brought costs of their own. Accepting cards needed a machine, and the shopkeeper paid a fee on every sale. Those costs shut out the small traders who make up most of Indian retail. Mobile wallets then appeared and made acceptance cheaper. But they created a worse problem. Each wallet was a closed loop. Money in one company's wallet could not reach a user of another. So the market split into networks that could not reach each other.

UPI's big idea was to treat the network itself as a public good. NPCI did not build a better wallet. Instead it wrote a standard that any wallet, bank or app could use, and made them all settle through the same system. The Bank for International Settlements calls designs like this digital public infrastructure. It means open systems that anyone can join and that serve the whole country. The BIS treats fast payment systems as the clearest example, whether the operator is public or private [20].

3.2 UPI and its features

Six features make UPI different from what came before. Together they explain the pattern of growth described later in this chapter.

- Interoperability. Any app can pay any account at any member bank. By FY 2025-26 there were 703 banks on the platform, up from 44 in FY 2016-17 [2].

- No account details needed. A UPI address or a QR code replaces the account number and branch code. This removes something to remember and a common cause of mistakes.
- Always available. Payment is instant and the system runs without a break, including on holidays.
- Free for the user. There is no charge to the payer. Small shopkeepers pay no Merchant Discount Rate on payments up to ₹2,000. The Government pays them 0.15 per cent of the value instead [14].
- Two-factor security built in. Every payment needs a PIN, and the app is tied to the customer's registered mobile number [28].
- Easy to extend. The same system now carries standing instructions, credit lines, payments abroad and UPI-linked credit cards. None of this changes what the user sees.

3.3 The growth of UPI transactions in India

The full growth record is in Data Set 1. Two points are worth making here, because they shape how the rest of this chapter should be read.

The first is that growth has kept going, rather than spiking and then flattening. A system that grows a thousand times in three years and then stops has found a small niche. A system that keeps growing for ten years has changed a habit. UPI volume rose in every single year I studied. That includes years long after the novelty had worn off.

The second is that volume has grown faster than value, year after year. This is not just a statistical oddity. It is the most useful fact in the whole data set, and Data Set 2 is about it. It means the typical UPI payment has been getting smaller. That tells us something about what kind of spending is moving onto the platform.

3.4 The concept of the velocity of money

Velocity is one of the oldest ideas in monetary economics. It is also one of the most often misused. I explain the theory carefully here, because my later claims depend on getting it right.

3.4.1 The equation of exchange

Irving Fisher's equation of exchange says that the total value of payments in an economy must equal the total value of what was exchanged:

$$M \times V = P \times T$$

Here M is the money stock, V is transactions velocity, P is the average price and T is the number of transactions. This is an identity. That means it is true by

definition, not because of any assumption. Its value is in what it makes us notice. If the money stock stays the same and the number of transactions rises, then velocity must have risen.

3.4.2 Income velocity and transactions velocity

We cannot observe T directly. So economists usually use the income form of the equation. In this version, total transactions are replaced by national income:

$$V = (P \times Y) \div M$$

This is income velocity, and it is the safer measure. It counts only payments made for newly produced goods and services. Transactions velocity counts every payment. That includes money moved between a person's own accounts, money paid back to a friend, and a wholesaler paying a distributor for goods that enter GDP only once, at the final sale.

The gap between the two is large, and it always runs the same way. A payment system that makes transfers easy will raise transactions velocity a lot and income velocity much less. This is because most of the extra payments are not purchases of final output. So if a study says a payment system raised "the velocity of money" without saying which one, it should be read with care. That is why I report a measure of transactions velocity in Data Set 4 and label it clearly.

3.4.3 The Cambridge approach and the demand for money

The Cambridge cash-balance approach says the same thing from the other side. It writes $M = k \times P \times Y$. Here k is the share of income that people choose to keep as money. Velocity is simply one divided by k . So a rise in velocity is exactly the same thing as a fall in the share of income kept idle as money.

This version makes the link to payment technology easy to see. Keynes said people hold money for transactions because income and spending do not arrive at the same time. They need a balance to bridge that gap. A system that lets them transfer money instantly, at any hour, from an account that earns interest makes the bridge smaller. So balances fall compared with spending. That means k falls and V rises. This is how UPI could affect velocity, and it is what I test in this project.

3.4.4 What velocity does not tell us

One warning is needed here. A rise in velocity is not by itself a sign of a healthy economy. Velocity rises sharply during hyperinflation, when holding money is punished and people spend at once. It falls in uncertain times, when people save for safety. Faster circulation is only a good sign when it comes from cheaper

payments, and not from people running away from money itself. This matters for how Data Set 4 should be read.

3.5 Digital payments and the circulation of money

The Reserve Bank treats the link between digital payments and the demand for cash as a policy matter, not just an academic one. Payments Vision 2025 lists cutting cash in circulation, as a share of GDP, among its goals. It notes that India's ratio is high compared with other countries. It also notes that the demand for cash grows with both output and inflation [26].

Digital payments do not cut the demand for cash by making people poorer. They change the form in which money is held. Money kept in a bank account, and spendable at once through a phone, does the job cash used to do. The difference is that it stays inside the banking system, where it can be lent out. The Reserve Bank tracks this shift with its Digital Payments Index. The index uses March 2018 as a base of 100. It stood at 465.33 in September 2024 and 493.22 in March 2025 [6].

Evidence on household behaviour points the same way. The Department of Financial Services study of February 2026 found that users of UPI and RuPay reported a clear fall in cash use and in ATM withdrawals. Of those users, 74 per cent said speed of payment was the main advantage [15].

3.6 Digital payments and economic activity

Research on whether fast payment systems affect real activity is still developing. It is also more careful than most popular writing on the subject.

The Bank for International Settlements studied 86,163 apps across 95 countries between 2012 and 2022. It found that launching a fast payment system increases the use of digital finance apps. The effect is largest in lower-income countries. It was strongest for apps built by new fintech and big-tech firms, rather than by older banks. It was also larger where the central bank took an active role, where anyone could join, and where settlement happened in real time [20]. India's system has all three of these features.

The International Monetary Fund studied interoperability using data on every UPI payment. It found that digital payments later grew fastest in the districts where the payment market had been most divided before. Those were the places where joining it together helped most [18]. This comes closer to showing cause and effect than most work in the field. It compares districts that received changes of different sizes, instead of just comparing before with after.

Evidence from similar economies points the same way. A study of thirty-three Indonesian provinces found that digital payments had a real effect on regional

income and consumption. This was true both before and after a structural break linked to COVID-19, and the effect was larger after the break [29]. Indonesia is not India, so this cannot simply be transferred. I quote it to show that this kind of data can detect the relationship at all, not as a measurement of India.

For India itself, the most useful evidence is about behaviour rather than national figures. One study surveyed 276 people and then interviewed 20 of them. About 75 per cent said they spent more because of UPI. Many said digital money felt less real, so they minded parting with it less. In the same study, 95.2 per cent found UPI convenient and 91.5 per cent were satisfied with it [25]. The sample is small and the answers are self-reported, so the results cannot be applied to the whole country. But the study shows something national figures cannot see. It also carries a warning as well as a finding. A system that makes spending easier also makes overspending easier.

The evidence from shopkeepers is stronger. The Department of Financial Services study found that 94 per cent of small merchants had adopted UPI. About 72 per cent were happy with digital payments, giving faster payment and better record-keeping as their reasons. Most important for this project, 57 per cent said their sales had risen after they started accepting digital payment [15]. These are reported figures, not audited accounts. A shopkeeper may credit a payment method for a gain that had another cause. Even so, a majority reporting higher sales is the most direct evidence we have that the effect reaches real activity, and not just the way people pay.

3.7 Financial inclusion and the digital economy

Financial inclusion in India was built in layers, and the payment layer made the others useful. A Jan Dhan account that is opened and never used is just an entry in a register. The same account, linked to a payment system that a vegetable seller will accept, becomes a working financial relationship.

The Reserve Bank's Financial Inclusion Index records this change. It rose from 64.2 in March 2024 to 67.0 in March 2025, and all three parts improved: access, usage and quality [7]. The rise in usage matters most. Access had already been largely achieved through the Jan Dhan scheme, which had reached over 55.98 crore people by August 2025. More than 55 per cent of those were women [7].

The World Bank's Global Findex 2025 surveyed about 145,000 adults in 141 countries. It is the international standard for measuring this. It records one Indian finding that matters here. The share of Indian women holding inactive accounts fell from a third in 2021 to 18 per cent by 2024 [21]. Inactive accounts are exactly the problem that a usable payment system solves.

Beyond payments, the record of transactions has itself become valuable. The Reserve Bank built the Unified Lending Interface on this idea. It gives approved lenders easy access to checked information about borrowers. This turns a payment history into proof that someone can repay a loan [20]. For a small trader who has never had audited accounts, this is how accepting digital payments can become access to credit.

3.8 Government initiatives promoting digital payments

Growth on this scale did not happen on its own. Four kinds of government action shaped it.

- **Pricing.** The Government has kept UPI free for users. It pays the system for the income it gives up, rather than letting shopkeepers be charged. Several schemes have paid banks a percentage of the payment value instead of a Merchant Discount Rate. One of these was a ₹1,500 crore scheme for small merchant payments [14], [15].
- **Infrastructure.** The Payments Infrastructure Development Fund pays part of the cost of putting payment infrastructure into smaller towns. By 31 October 2025 about 5.45 crore digital payment points had been set up through the Fund in tier-3 to tier-6 centres. By FY 2024-25, about 56.86 crore QR codes had been given to around 6.5 crore merchants [14].
- **Identity and accounts.** The Jan Dhan, Aadhaar and Mobile combination gave people what they needed first. That was an account to receive money, an identity to prove who they were, and a phone to pay from.
- **Going international.** Through NPCI International Payments Limited, UPI payments and money transfers now reach eleven foreign countries. Acceptance began in Cambodia in June 2026. A money transfer link with the Maldives' instant payment system started in July 2026 [17].

The cost of this policy should be stated honestly. Free payments are not costless payments. The cost has simply moved from the shopkeeper to the Government. Whether that is worth it depends on whether the gains in recorded activity, inclusion and tax collection are bigger than the amount spent. I raise this question in Chapter 5 but cannot answer it.

3.9 Challenges faced by digital payment systems

Four problems come up again and again in the research and in policy papers. An honest assessment has to give them proper weight.

Cyber fraud and tricking users.

Cybersecurity incidents in India rose from 10.29 lakh in 2022 to 22.68 lakh in 2024 [28]. The technology itself is not the main weak point. Tying the app to one device and asking for a PIN make it hard to break in. The weak point is the user, who can be tricked into approving a payment. NPCI runs a fraud-monitoring system that uses machine learning to raise alerts and block suspicious payments. The Citizen Financial Cyber Fraud Reporting and Management System has saved more than ₹1,200 crore across more than 4.7 lakh complaints [28]. A system used by over 55 crore people attracts attacks to match its size.

Dependence on infrastructure.

A digital payment needs electricity, a working phone and a network signal, all at the same moment. Cash needs none of these. So as UPI replaces cash, the risk becomes concentrated. A breakdown that was once an inconvenience now stops trade.

Unequal access.

Data Set 8 measures the digital divide. In August 2025 rural areas had 46.73 internet subscribers per 100 people. Urban areas had 113.83 [22]. A payment system that can only be reached through the internet cannot include more people than the internet does.

Risk to spending habits.

About 75 per cent of surveyed users said they spent more because digital money felt less real [25]. That is a real cost. It matters most for low-income families, who have little room to absorb overspending. The same ease that raises velocity also removes a restraint that cash used to provide by accident.

3.10 Case studies

3.10.1 Urban and rural India

The rural story is one of fast catching up from a low start. This was done by deliberately paying for payment infrastructure, rather than waiting for the market. The Payments Infrastructure Development Fund exists for this reason. Putting a QR code in a tier-6 town is not profitable at the small volumes available there at first. By October 2025 the Fund had set up 5.45 crore payment points in tier-3 to tier-6 centres [14]. But the gap that remains is not mainly about acceptance. It is about connectivity, and Data Set 8 measures it.

3.10.2 Small businesses and consumers

For the small shopkeeper, UPI solved a problem that cards never could. A card machine costs money, and a fee on every sale eats into a thin margin. A printed

QR code costs nothing, and below ₹2,000 there is no fee at all [14]. So almost everyone accepts it. Of the small merchants surveyed, 94 per cent had adopted UPI and 57 per cent said their sales rose afterwards [15]. Street vendors under the PM SVANidhi scheme are part of the same system. They get cashback on digital payments and access to UPI-linked RuPay credit cards, which turn a payment record into a line of credit [14].

3.10.3 India in international comparison

India stands out in two ways. The first is size. UPI handled nearly 49 per cent of all real-time payments made in the world in 2024. The IMF calls it the largest retail fast payment system by volume [3], [18]. The second is pricing. Most similar systems abroad charge shopkeepers something. The BIS notes that in India there is no charge to the user for UPI. A fee applies only to merchant payments made through prepaid instruments like wallets [20]. India chose to treat retail payment as a public service. The growth described in this chapter is partly a result of that choice.

4. Analysis of Data

This chapter has the main analysis of my project. I look at eight data sets. Each one follows the same four-part pattern: the table or graph, an explanation of what the data show, an analysis of what they mean, and an interpretation of what can and cannot be concluded. There are ten figures and eight tables. Each data set ends with one sentence summing up the result.

The first four data sets look at growth, size of payments and velocity. Data Sets 5 and 6 test the two big shocks of the period. Data Sets 7 and 8 look at financial inclusion and its limits. Together they test the three hypotheses from Section 2.2.

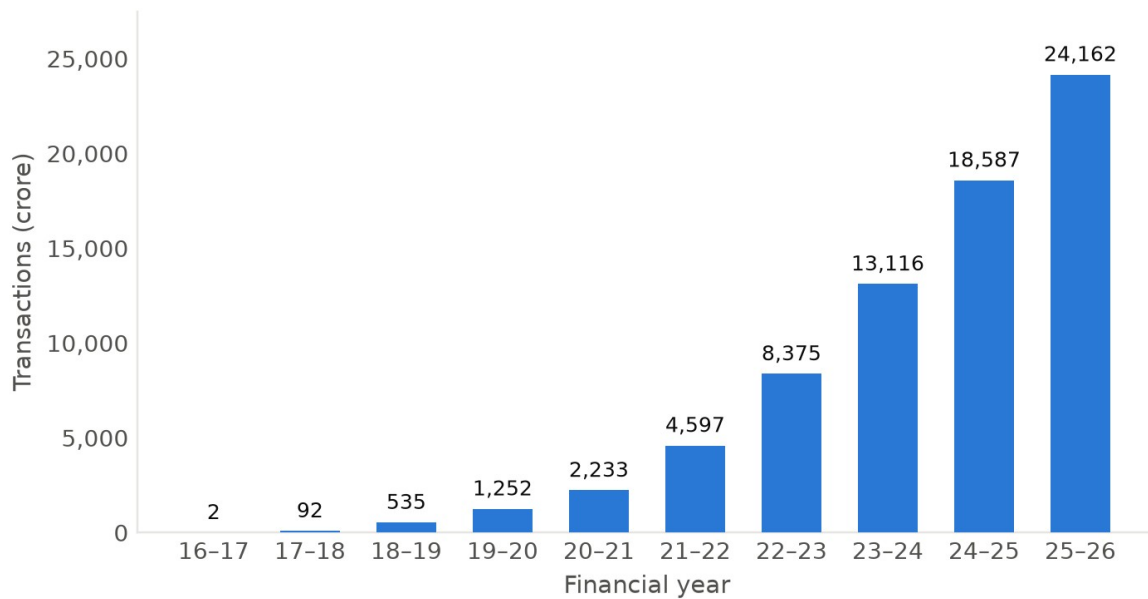
4.1 Data Set 1 · The growth of UPI transactions

Financial year	Volume (crore)	Value (₹ lakh crore)	Volume growth	Value growth
2016-17	2	0.07	—	—
2017-18	92	1.10	4,500%	1,471%
2018-19	535	8.77	482%	697%
2019-20	1,252	21.32	134%	143%
2020-21	2,233	41.04	78%	92%
2021-22	4,597	84.16	106%	105%
2022-23	8,375	139.00	82%	65%
2023-24	13,116	199.90	57%	44%
2024-25	18,587	261.00	42%	31%
2025-26	24,162	314.00	30%	20%

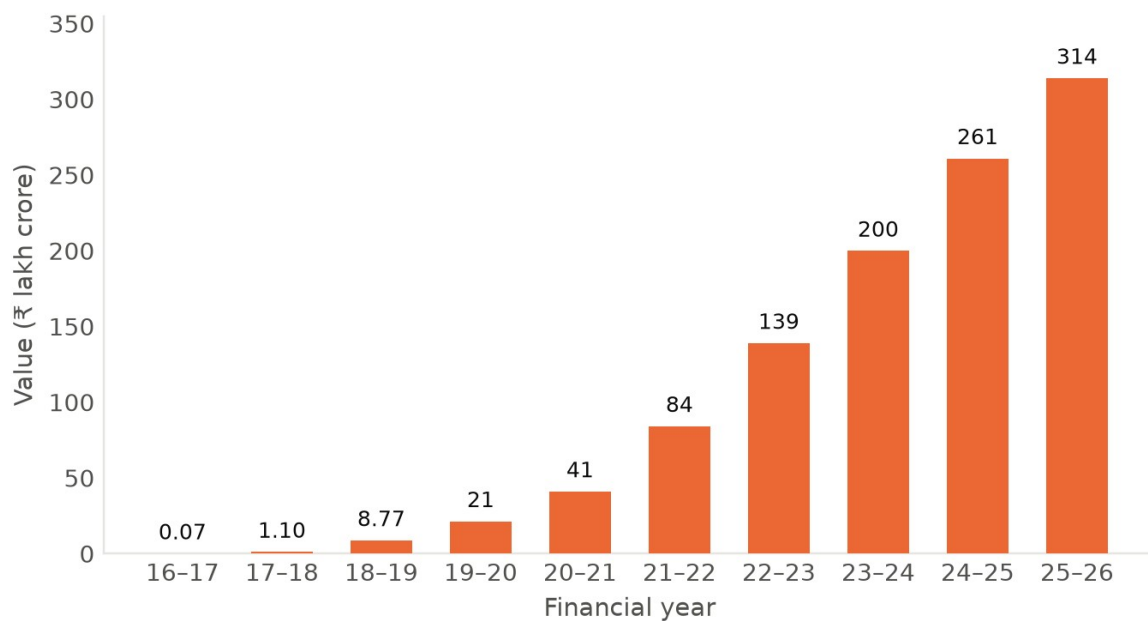
Table 1. Sources: NPCI UPI product statistics [1], with year-wise figures as reported in Press Information Bureau releases [2], [10], [13], [16]. Growth rates are computed by the author from the volume and value columns. Figures are rounded as published.

Figure 1 UPI transaction volume, FY 2016-17 to FY 2025-26

Annual number of UPI transactions, in crore

*Figure 1. UPI transaction volume, FY 2016-17 to FY 2025-26.***Figure 2 UPI transaction value, FY 2016-17 to FY 2025-26**

Annual value of UPI transactions, in ₹ lakh crore

*Figure 2. UPI transaction value, FY 2016-17 to FY 2025-26.***Explanation of the data**

The table shows the number and value of UPI payments in each financial year since UPI started. It also gives the growth rates, which I worked out from those

two columns. Volume rose from 2 crore payments to 24,162 crore. Value rose from ₹0.07 lakh crore to about ₹314 lakh crore. Both of these happened over ten years [2].

Analysis

Three patterns stand out. The first is that the growth rate is slowing in percentage terms, from 4,500 per cent down to 30 per cent. But the actual rise each year keeps getting bigger. The rise in FY 2025-26 alone was about 5,575 crore payments. That is more than the whole of FY 2021-22. This is simply what happens when a base gets larger. It does not mean UPI is really slowing down.

The second is that the growth never flattens out. Volume rose for ten years in a row, through a pandemic and long after the novelty wore off. That points to a lasting change in how people pay, not a passing trend.

The third pattern matters most. Volume has grown faster than value in every year since FY 2021-22. In FY 2025-26 volume grew 30 per cent while value grew 20 per cent. The average size of a payment is value divided by volume. So if volume grows faster, the average payment must be getting smaller. Data Set 2 looks at this directly.

Interpretation

UPI has gone from an experiment to the normal way India pays in shops. It now handles most retail digital payments. The gap between volume growth and value growth is the first sign of something more. It suggests UPI is not just taking over payments that already existed. It is reaching very small payments that used to be made in cash, or were not made at all. So H1 is supported as a description. Its effect on velocity is tested in Data Set 4.

In one sentence: UPI grew from 2 crore payments to 24,162 crore in ten years. Volume grew faster than value in every year from FY 2021-22, so the typical payment was getting smaller even as the total kept rising.

4.2 Data Set 2 · The average value of a single UPI transaction

Financial year	Value (₹ lakh crore)	Volume (crore)	Average per transaction (₹)
2017-18	1.10	92	1,196
2018-19	8.77	535	1,639
2019-20	21.32	1,252	1,703

Financial year	Value (₹ lakh crore)	Volume (crore)	Average per transaction (₹)
2020-21	41.04	2,233	1,838
2021-22	84.16	4,597	1,831
2022-23	139.00	8,375	1,660
2023-24	199.90	13,116	1,524
2024-25	261.00	18,587	1,404
2025-26	314.00	24,162	1,300

Table 2. The average value per transaction is derived by the author as transaction value divided by transaction volume, using the figures in Table 1. FY 2016-17 is omitted because the base is too small for the ratio to be meaningful. Underlying data: [1], [2].

Figure 3 Average value of a single UPI transaction

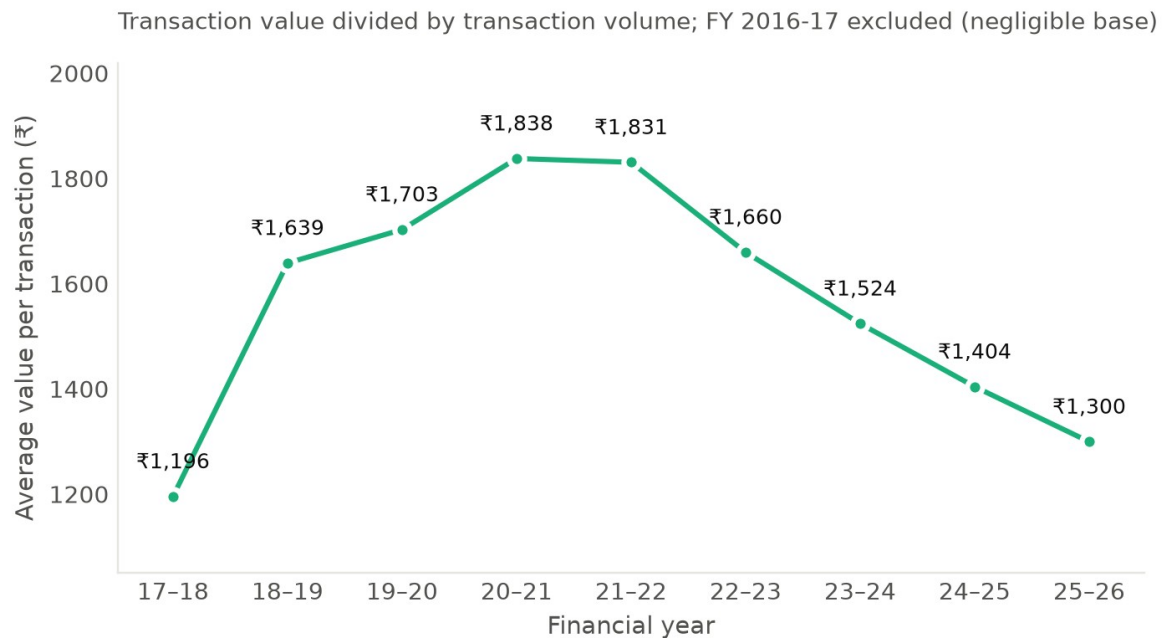


Figure 3. Average value of a single UPI transaction, FY 2017-18 to FY 2025-26.

Explanation of the data

This figure is not published anywhere. I worked it out by dividing each year's value by its volume. It answers a simple question: how big is a typical UPI payment, and is that changing? The answer rises from ₹1,196 in FY 2017-18 to a peak of ₹1,838 in FY 2020-21. After that it falls steadily to ₹1,300 in FY 2025-26. That is a drop of about 29 per cent from the peak.

Analysis

The shape of this curve shows UPI's growth in two clear phases.

In the first phase, up to FY 2020-21, the average payment grew. This is what you would expect when the early users are better off, live in cities and are used to technology. The payments moving onto UPI then were large ones, such as rent, fees and money sent to family. These used to be paid by cheque or bank transfer.

In the second phase the average falls, every year, for five years. Total value kept rising all through. So this cannot mean people were spending less. It can only mean the new payments being added were much smaller than the old average, and that there were enough of them to pull the average down. The data on payment types confirms this. Person-to-merchant payments are now 63 per cent of all UPI volume, and 86 per cent of those are below ₹500 [2].

Inflation works against this finding rather than causing it. Prices rose over the period. So the same basket of goods would show a rising average payment. The average fell anyway. That means the shift towards small payments is even bigger in real terms than it looks.

Interpretation

I think this is the most important finding in my project. It separates the two possible explanations of UPI's growth. If UPI had only digitised payments that already existed, the average payment would have stayed roughly flat while volume rose. Instead it fell by nearly a third while volume grew thirteen times. That is the sign of a payment system reaching payments it never served before: the tea, the vegetables, the auto fare.

This matters directly for velocity. Fisher's equation links the money stock to the number of transactions it supports. A system that adds a very large number of very small payments raises T a lot, while adding fairly little to the total value. That is exactly what it means for each rupee to be used more often. So H2 is supported.

In one sentence: The average UPI payment fell from ₹1,838 to ₹1,300 while volume rose more than ten times. That is the sign of a system adding small everyday payments, not just digitising ones that already existed.

4.3 Data Set 3 · UPI within India's wider digital payment system

Indicator	Earlier period	Later period
Total digital payment volume (crore)	2,071 (FY 2017-18)	18,737 (FY 2023-24)
Total digital payment value (₹ lakh crore)	1,962 (FY 2017-18)	3,659 (FY 2023-24)

Indicator	Earlier period	Later period
Volume CAGR over the period	—	44%
Value CAGR over the period	—	11%
UPI share of retail digital payment volume	—	81% (FY 2024-25)
Digital transactions, FY 2019-20 to FY 2024-25	—	over 65,000 crore, worth over ₹12,000 lakh crore

Table 3. Sources: Press Information Bureau releases [10], [11], [12], [14].

Figure 5 UPI's share of retail digital payment volume, FY 2024-25

Share of the total number of retail digital payment transactions in India

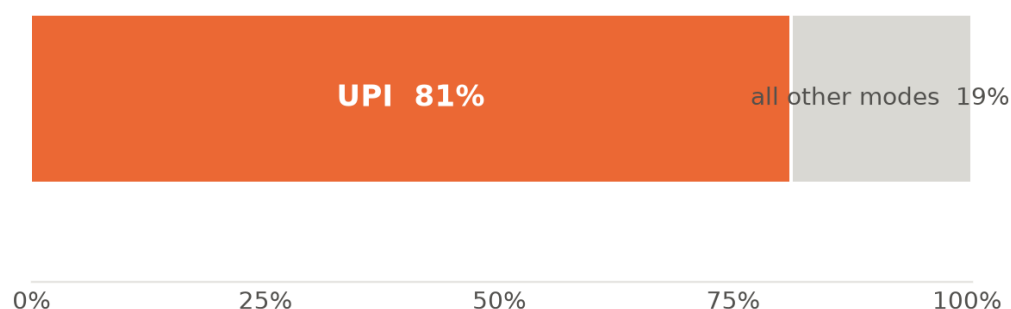


Figure 5. UPI's share of retail digital payment volume, FY 2024-25.

Explanation of the data

This data set puts UPI inside all digital payments in India. That includes cards, wallets, NEFT, IMPS and RTGS. Total digital payment volume rose from 2,071 crore payments in FY 2017-18 to 18,737 crore in FY 2023-24. That is a compound growth rate of 44 per cent a year. Value grew from ₹1,962 lakh crore to ₹3,659 lakh crore, a compound rate of 11 per cent [10], [11]. UPI alone made up 81 per cent of retail digital payment volume in FY 2024-25 [14].

Analysis

The gap between those two growth rates is 44 per cent for volume against 11 per cent for value. This repeats, across the whole payment system, the same pattern I found in UPI alone. India's digital payments have grown mainly because more payments are being made, not because payments are bigger.

The value figures need care. Total digital payment value of ₹3,659 lakh crore is far bigger than India's GDP. That is because it includes RTGS, which carries very large transfers between banks and big companies. There are not many of these, but each one is huge. This shows clearly why payment value cannot be treated as

economic output. It is also why the ratio in Data Set 4 has to be called a transactions measure and not income velocity.

There is no contradiction in UPI carrying 81 per cent of retail payments by number while total value is dominated by large wholesale systems. It just describes what UPI is. It is the system for frequent, small, everyday payments.

Interpretation

UPI is not one choice among several in Indian retail payments. It is the system, and everything else shares the remaining fifth. For this study, that means changes in how India pays can fairly be linked to UPI, because nothing else is big enough to move the national figures.

In one sentence: UPI carries 81 per cent of India's retail digital payments. So changes in how people pay during this period can fairly be linked to it.

4.4 Data Set 4 · UPI turnover relative to nominal GDP

Financial year	UPI value (₹ lakh crore)	Nominal GDP (₹ lakh crore)	UPI ÷ GDP
2022-23	139.00	269.50	0.52×
2023-24	199.90	295.36	0.68×
2024-25	261.00	318.07	0.82×
2025-26	314.00	346.36	0.91×

Table 4. UPI values from [1], [2]. Nominal GDP at current prices from MoSPI provisional estimates [8], [9]. The ratio is computed by the author. Note a discontinuity: figures for FY 2022-23 and FY 2023-24 are on the 2011-12 base series, while FY 2024-25 and FY 2025-26 are on the revised 2022-23 base series. The break is small relative to the trend but the series is not perfectly continuous.

Figure 4 UPI turnover relative to nominal GDP

A transactions-velocity proxy: rupees moved over UPI per rupee of national output

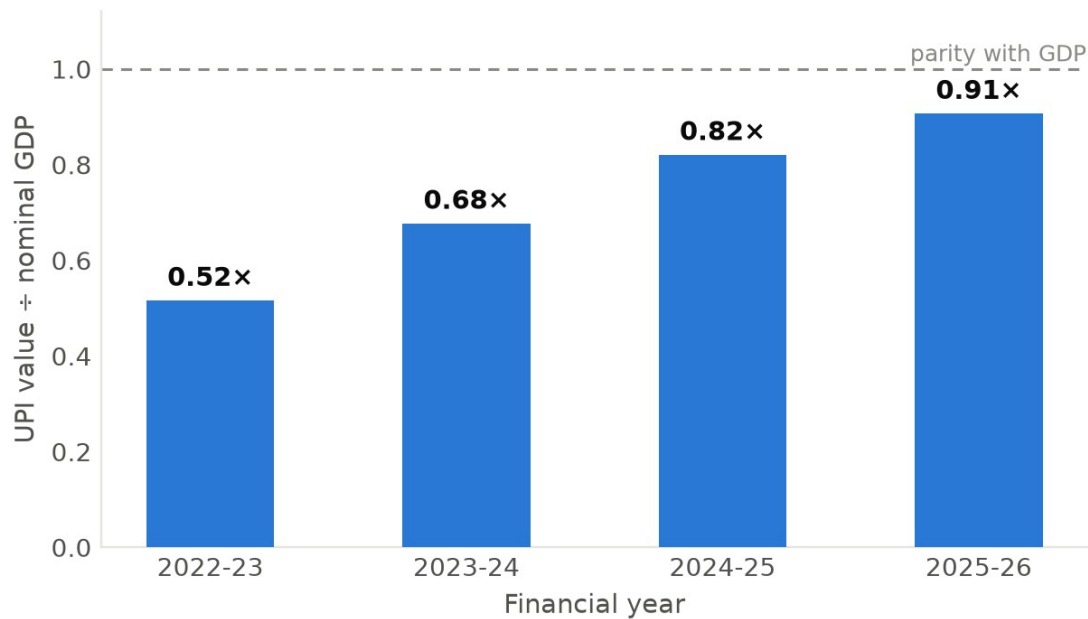


Figure 4. UPI turnover relative to nominal GDP, FY 2022-23 to FY 2025-26.

Explanation of the data

This data set builds the closest thing to a velocity measure that public data allows. It shows the total value settled over UPI in a year as a multiple of nominal GDP in the same year. The ratio rose from 0.52 in FY 2022-23 to 0.91 in FY 2025-26. By the latest year, the value passing through UPI alone came close to the whole yearly output of the Indian economy.

Analysis

I need to say exactly what this ratio does and does not measure. It would be easy to claim too much from it.

It is a measure of transactions turnover, not income velocity. The top of the ratio counts every payment. That includes transfers between people, which do not buy anything new, and payments between businesses, which GDP counts only once at the final sale. The bottom counts final output only. The two are not measuring the same thing. So the number should not be read as "each rupee now circulates 0.91 times".

What the ratio does measure reliably is the direction and speed of change. The mismatch between the top and the bottom stays roughly the same each year. So comparing one year with another is still fair. On that basis the ratio rose by about 75 per cent in three years. The value settled through this one retail system grew far faster than national output. So payment activity per unit of output has clearly

gone up. This is what the Cambridge version predicts. As paying gets cheaper, people hold less money against a given amount of spending, so turnover rises.

Two cautions are needed. First, part of the rise is just cash payments moving onto UPI. That raises the measured figure without any change in behaviour, because cash payments were never counted in the first place. Second, as Section 3.4.4 explained, rising velocity is only a good sign when it comes from cheaper payments rather than people running away from money. Here the cause is clearly the first one. India had no monetary crisis in this period, and the rise matches the fall in the cost and difficulty of paying.

Interpretation

H1 is supported, with one qualification. What I have shown is a rise in measured transactions turnover, not in income velocity. Part of that rise is cash payments becoming visible for the first time, rather than new activity. Both effects are real and both matter, but they are not the same thing. The published data cannot separate them. This is the honest limit of what a study based on national figures can show.

In one sentence: The value settled over UPI rose from 0.52 to 0.91 times India's GDP in three years. The level of that ratio is not meaningful, but its fast rise is, and it points to a higher transactions velocity of money.

4.5 Data Set 5 · Digital payments around demonetisation

Indicator	Value
Value of specified bank notes demonetised (Nov 2016)	₹15.4 trillion
Share of total value of notes in circulation	86.9%
Share of demonetised notes returned to the RBI	approximately 98.96%
Monthly digital transactions, October 2016	71.27 crore
Monthly digital transactions, May 2017	111.45 crore
Increase over the period	56%

Table 5. Sources: RBI, *Macroeconomic Impact of Demonetisation* [23]; PIB [24].

Figure 6 Digital payments around demonetisation

Monthly digital transaction volume, October 2016 vs May 2017

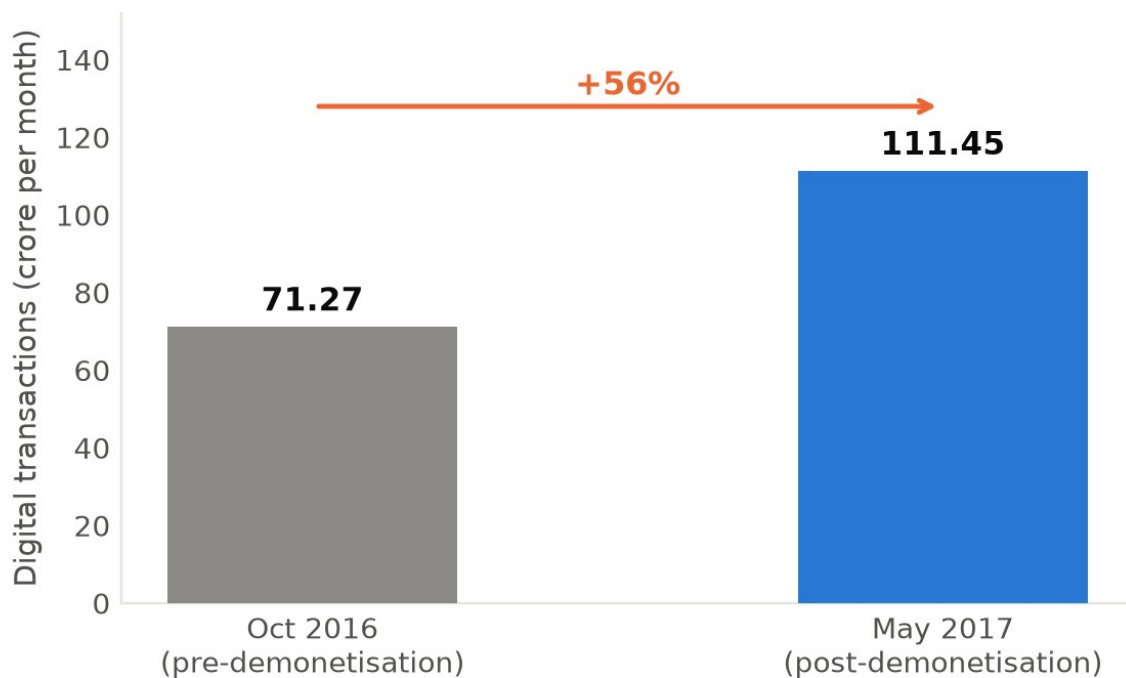


Figure 6. Monthly digital transaction volume before and after demonetisation.

Explanation of the data

On 9 November 2016, ₹500 and ₹1,000 notes stopped being legal tender. They were worth ₹15.4 trillion. That was 86.9 per cent of the value of all notes then in circulation [23]. Monthly digital payments rose from 71.27 crore in October 2016 to 111.45 crore by May 2017. That is a rise of 56 per cent [24].

Analysis

Demonetisation worked like a very severe natural experiment. The main way people paid was taken away almost overnight, and an alternative had launched only weeks before. The 56 per cent rise in digital payments over seven months is how people reacted.

That reaction should not be overstated. About 98.96 per cent of the withdrawn notes came back to the banking system [23]. So the currency was replaced rather than removed for good, and cash use recovered a lot in the years that followed. Demonetisation did not by itself turn India into a digital economy.

What it did achieve is better described as a forced trial. Many people who would not otherwise have installed a payment app did so because they had to. Many shopkeepers who would not otherwise have accepted digital payment learned how. Some of that behaviour stayed once the pressure was gone. So the lasting

effect was on familiarity and infrastructure, rather than on how people actually paid.

Interpretation

Demonetisation speeded up UPI's growth at an early and important moment. But it did not cause the growth that followed. That growth carried on for years after cash supply returned to normal, and at rates far above anything seen in 2016-17. So the event is better understood as removing a barrier, which was the effort of learning something new. It did not change the economics of paying.

In one sentence: Demonetisation produced a 56 per cent jump in digital payments. But about 99 per cent of the currency came back, so it speeded up UPI's growth without causing it.

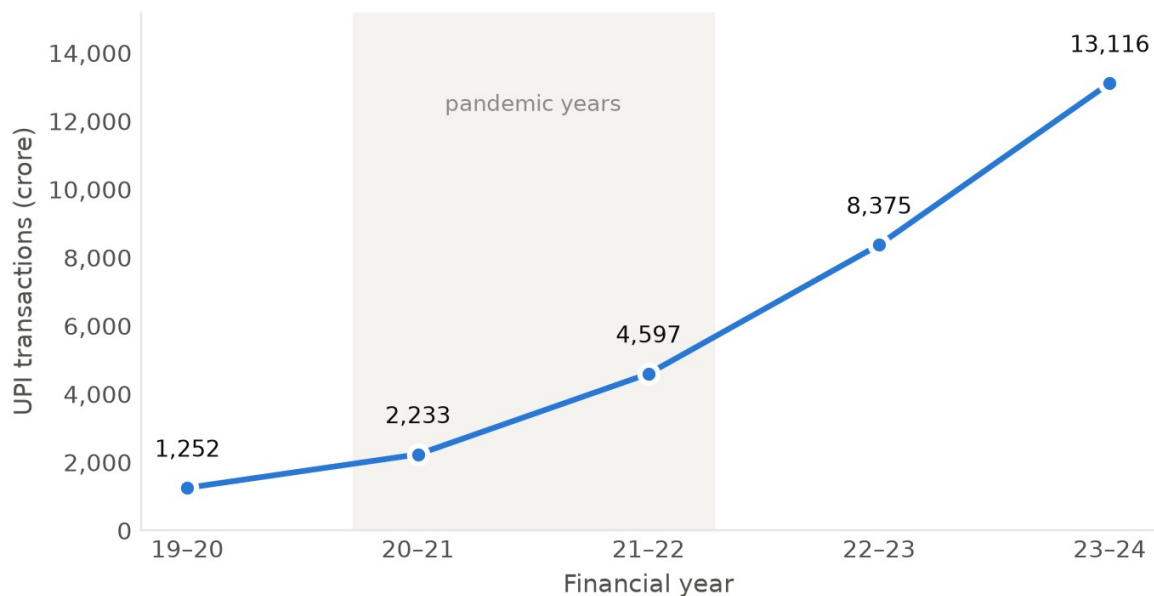
4.6 Data Set 6 · UPI growth before, during and after COVID-19

Financial year	Volume (crore)	Growth	Period
2019-20	1,252	134%	Pre-pandemic
2020-21	2,233	78%	Pandemic
2021-22	4,597	106%	Pandemic
2022-23	8,375	82%	Post-pandemic
2023-24	13,116	57%	Post-pandemic

Table 6. Volumes from [1], [2], [10], [13]. Growth rates computed by the author.

Figure 7 UPI volume before, during and after COVID-19

FY 2019-20 to FY 2023-24; shaded band marks FY 2020-21 and FY 2021-22

*Figure 7. UPI transaction volume before, during and after the COVID-19 pandemic.***Explanation of the data**

This table separates out the pandemic years. UPI volume grew 78 per cent in FY 2020-21, the year of the strictest restrictions. It then grew 106 per cent in FY 2021-22, which is a higher rate than the year before. Volume more than tripled between FY 2019-20 and FY 2021-22.

Analysis

The pattern looks surprising at first. India's real output shrank in FY 2020-21. You would expect a payment system to shrink along with the economy it serves. Instead UPI volume grew 78 per cent. Two forces explain this.

The first is substitution. With movement restricted and contact discouraged, payments that would have been made in cash were made digitally instead. This moves payments between methods. It does not increase how many there are.

The second is new users, and this matters more. Shopkeepers who had refused digital payment accepted it because customers insisted. Customers who had avoided it learned because there was no other option. This permanently widened the number of people who could use the system.

The proof that new users mattered more lies in what happened next. If the pandemic rise had been pure substitution, growth would have reversed once normal trade came back. Instead FY 2021-22 grew faster than FY 2020-21.

Growth then carried on at 82 per cent and 57 per cent in the next two years, from a much bigger base. The change did not unwind.

Interpretation

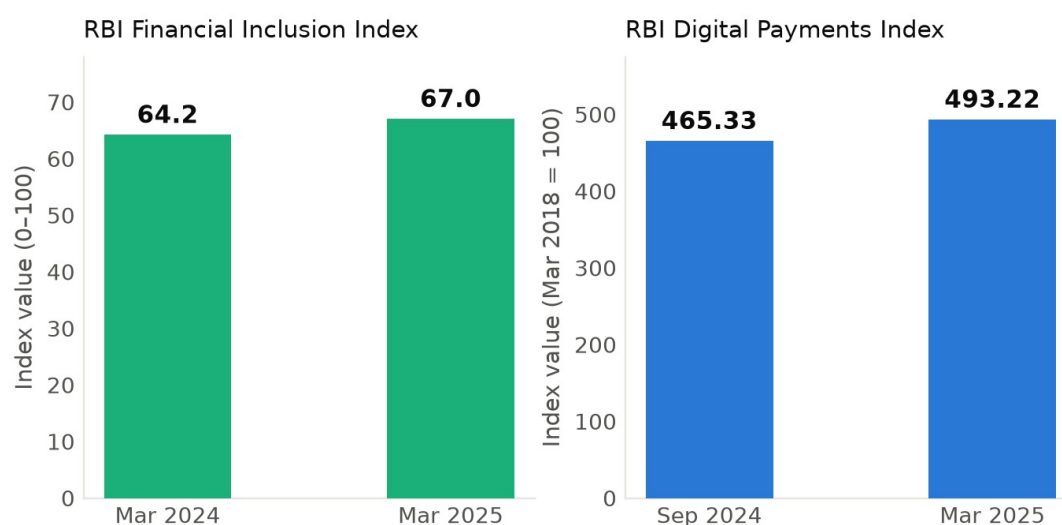
COVID-19 was a second shock, and a more lasting one than demonetisation. The contrast between the two is useful. Demonetisation removed the alternative for a few months, and the change partly reversed afterwards. The pandemic lasted long enough for digital payment to become a habit. Habits, unlike forced behaviour, survive once the pressure is removed.

In one sentence: UPI grew faster after the pandemic (106 per cent) than during its worst year (78 per cent). That shows the change had become a habit, instead of reversing once cash was available again.

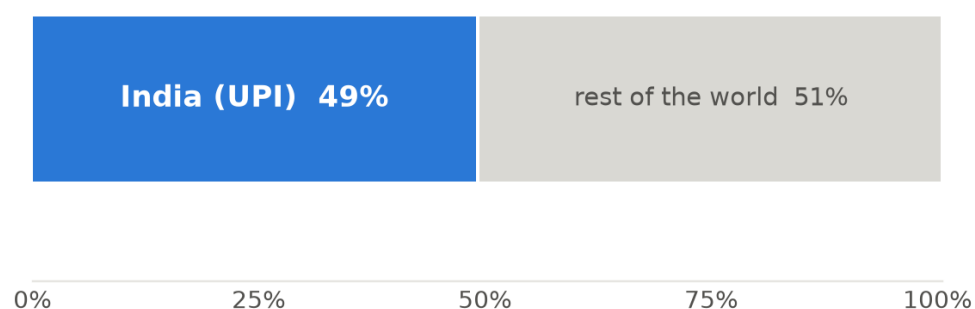
4.7 Data Set 7 · Financial inclusion, digitalisation and global standing

Indicator	Earlier	Later
RBI Financial Inclusion Index	64.2 (Mar 2024)	67.0 (Mar 2025)
RBI Digital Payments Index (Mar 2018 = 100)	465.33 (Sep 2024)	493.22 (Mar 2025)
PMJDY beneficiaries	—	55.98 crore (Aug 2025)
Share of PMJDY accounts held by women	—	over 55%
UPI users onboarded	—	55.49 crore (Jun 2026)
Merchants accepting UPI	—	approx. 6.5 crore (FY 2024-25)
QR codes deployed	—	56.86 crore (FY 2024-25)
India's share of global real-time payment volume	—	approx. 49% (2024)

Table 7. Sources: RBI and PIB releases [3], [6], [7], [14], [16]; IMF [18].

Figure 8 Financial inclusion and payment digitalisation, RBI indices*Figure 8. RBI Financial Inclusion Index and RBI Digital Payments Index.***Figure 10 India's share of global real-time payment volume, 2024**

As recognised by the IMF, June 2025

*Figure 10. India's share of global real-time payment volume, 2024.***Explanation of the data**

This data set brings together the main signs of financial inclusion and digitalisation. Both RBI indices rose over their latest periods. The Financial Inclusion Index improved in all three of its parts: access, usage and quality [7]. The last row shows that India handled about 49 per cent of all real-time payments in the world in 2024 [3].

Analysis

The rise in the usage part matters more than the headline number. Access had already been largely achieved through the Jan Dhan scheme. The real problem was accounts lying unused. The World Bank records that the share of Indian women with inactive accounts fell from a third in 2021 to 18 per cent by 2024

[21]. A payment system that a small vendor will accept is what turns an account from a registration into something useful.

The international comparison needs to be kept in proportion. India's 49 per cent share of world real-time payments partly reflects our population. A share of volume also says nothing about quality. What makes it notable is the policy difference the BIS points out. Most fast payment systems abroad charge shopkeepers, and India's does not [20]. India chose to treat retail payment as a public service, and this growth is partly a result of that choice.

It is worth noticing how close 55.98 crore Jan Dhan accounts and 55.49 crore UPI users are. These are different groups measured on different dates, so they cannot simply be set against each other. But their similar size suggests that owning an account and being able to pay digitally now reach a similar share of adults.

Interpretation

The gains in financial inclusion are real and can be measured. The chain running from accounts, to usage, to activity being recorded is visible in the indices rather than just claimed. India's international position rests on a deliberate choice about pricing, not on size alone.

In one sentence: The RBI's Financial Inclusion Index rose from 64.2 to 67.0, with usage improving fastest. That is the part that matters, because access had already been achieved and unused accounts were the real problem.

4.8 Data Set 8 · Internet penetration and the rural-urban divide

Indicator	Urban	Rural
Internet subscribers, Aug 2025 (million)	579.46	423.39
Internet subscribers per 100 people, Aug 2025	113.83	46.73
Tele-density, Dec 2025	148.92%	59.63%
Telecom subscribers, Dec 2025 (million)	762.44	543.70

Table 8. Source: TRAI telecom subscription data and performance indicator reports [22]. Total broadband subscribers stood at 999.81 million as at 31 October 2025, and total telecom subscribers at 1,306.14 million as at 31 December 2025, giving an overall tele-density of 91.74 per cent.

Figure 9 The rural-urban digital divide

Urban internet penetration is roughly 2.4 times the rural rate

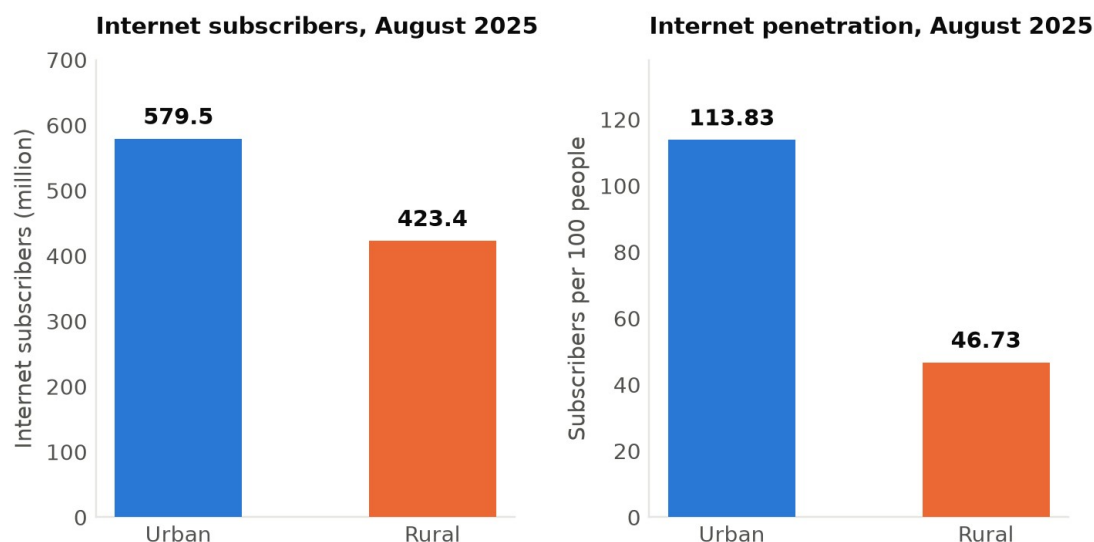


Figure 9. Internet subscribers and internet penetration, rural and urban India.

Explanation of the data

UPI needs a connected device. So internet access sets the limit on how inclusive it can be. India had 999.81 million broadband subscribers on 31 October 2025 [22]. Urban areas had 113.83 internet subscribers per 100 people. Rural areas had 46.73. That is a ratio of roughly 2.4 to 1.

Analysis

The urban figure is above 100 per 100 people because it counts subscriptions, not people. Many urban users have more than one. This is a reminder to read the rural figure as an upper limit. If some rural users also have more than one subscription, then fewer than 46.73 per cent of rural people are actually online.

In absolute numbers the gap is smaller than the penetration figures suggest. There are 579.46 million urban subscribers against 423.39 million rural ones. That is because rural India has more people. So the two panels of Figure 9 tell different halves of the same story. Rural India is a large and growing market in total, but a much smaller share of rural people is reached.

This is a limit of connectivity, not of acceptance. The Payments Infrastructure Development Fund has set up about 5.45 crore digital payment points in tier-3 to tier-6 centres [14]. So the problem is not that rural shops lack QR codes. The problem is that many rural people cannot reliably reach the network those codes need.

Interpretation

H3 is supported. The benefits of UPI described in this chapter are shared unevenly. The unevenness is built into the infrastructure rather than being a matter of choice, because it follows the spread of connectivity. So any policy meant to take digital payment further into rural India has to deal with the telecom problem. Payment measures alone cannot solve it.

In one sentence: Rural internet reaches 46.73 per 100 people against 113.83 in urban areas. So what limits UPI in rural India is connectivity, not whether shops will accept it.

5. Findings and Suggestions

5.1 Findings

Seven findings come out of the analysis in Chapter 4. I give each one with the evidence behind it, and with the qualification that evidence needs.

Finding 1. UPI has become the normal way India pays in shops.

Volume rose from 2 crore payments in FY 2016-17 to 24,162 crore in FY 2025-26. UPI made up 81 per cent of retail digital payments in FY 2024-25 [2], [14]. Growth happened in every single year. The actual rise each year keeps getting bigger, even though the percentage growth is slowing.

Finding 2. The growth is made up of small payments, more and more so.

The average UPI payment fell from ₹1,838 in FY 2020-21 to ₹1,300 in FY 2025-26. That is a fall of about 29 per cent. Over the same years volume rose more than ten times (Data Set 2). Person-to-merchant payments are now 63 per cent of volume, and 86 per cent of those are below ₹500 [2]. Prices rose over this period, so the shift towards small payments is even bigger in real terms than the figures show.

Finding 3. The value of payments settled for each rupee of output has risen sharply.

UPI value as a multiple of nominal GDP rose from 0.52 in FY 2022-23 to 0.91 in FY 2025-26 (Data Set 4). This fits the Cambridge prediction. When paying gets cheaper, people hold less money against a given amount of spending. But it is a transactions measure, not income velocity. Part of the rise is cash payments becoming visible for the first time, and the data cannot separate the two.

Finding 4. Both shocks speeded up growth, but only one lasted.

Digital payments rose 56 per cent in the seven months after demonetisation [24]. But about 98.96 per cent of the withdrawn currency came back to the banks [23], and cash use recovered a lot. The pandemic produced 78 per cent growth in FY 2020-21 and then 106 per cent in FY 2021-22. That is faster after the shock than during it. Growth then carried on at 82 and 57 per cent in the next two years (Data Set 6). The difference seems to be how long each one lasted. The pandemic went on long enough for digital payment to become a habit.

Finding 5. UPI has helped financial inclusion in ways we can measure.

The RBI Financial Inclusion Index rose from 64.2 to 67.0 between March 2024 and March 2025. Access, usage and quality all improved [7]. The gain in usage matters most, because access had already been largely achieved. The share of

Indian women with inactive accounts fell from a third in 2021 to 18 per cent by 2024 [21].

Finding 6. There is believable evidence of an effect on real activity, but it is not proof.

Of the small shopkeepers surveyed for the Department of Financial Services, 94 per cent had adopted UPI. Of those, 57 per cent said their sales had risen afterwards [15]. These are reported figures that cannot be audited, and a shopkeeper may credit a payment method for a gain that had another cause. Even so, it is the most direct evidence we have that the effect reaches output, and not just the way people pay. Research from other countries points the same way. The BIS finds that launching a fast payment system increases the use of digital finance, most of all in lower-income countries [20].

Finding 7. The gains are shared unevenly, and two serious risks come with them.

In August 2025 rural areas had 46.73 internet subscribers per 100 people, against 113.83 in urban areas [22]. That sets a ceiling on how inclusive a system needing a network can be. Cybersecurity incidents rose from 10.29 lakh in 2022 to 22.68 lakh in 2024 [28]. And about 75 per cent of users in one survey said they spent more because digital money felt less real [25]. The same ease that raises velocity also removes a restraint that cash used to give by accident.

5.2 Suggestions

Six suggestions follow from these findings. Each one deals with a problem the evidence actually shows, rather than a general wish.

17. Treat rural connectivity as payments policy. Data Set 8 shows that what limits rural digital payment is network access, not whether shops will accept it. About 5.45 crore payment points have been set up in tier-3 to tier-6 centres [14]. Yet fewer than half of rural people are connected. So spreading broadband would do more good now than paying for still more acceptance infrastructure.
18. Spend on fraud awareness on the same scale as the system itself. UPI's technical defences are strong. The app is tied to one device, a PIN is needed, and machine learning watches for suspicious payments [28]. The weak point is the user's judgement. So awareness campaigns should be treated as essential infrastructure. They should be in regional languages, inside the payment apps, at the moment of paying.
19. Build spending-awareness features into payment apps. Most surveyed users say they spend more because digital money feels less real [25]. Apps

could offer optional monthly spending summaries, and limits that users set for themselves. This deals with a real cost without taking away anyone's choice.

20. Publish payment data region by region. The biggest limit I faced was that UPI data is not published by state or district. This made it impossible to study regional differences. Releasing this data, with names removed, would allow research that national totals cannot support.
21. Reduce single points of failure. As cash disappears, a breakdown stops being an inconvenience and starts stopping trade. Offline payment modes, and modes that work on basic phones, deserve more investment. They should be funded so the system can survive failures, and not only to include more people.
22. Review the incentive scheme openly. Free payment is paid for by the Government rather than by shopkeepers [14], [15]. Whether the gains in recorded activity, inclusion and tax collection are worth that spending is a question that can be answered with evidence. It should be reviewed publicly from time to time, because the answer decides whether free payment can continue.

6. Conclusion

I set out to ask whether the growth of UPI has changed the speed at which money moves around in India. I also asked whether any effect on real economic activity can be found in published data. The two questions deserve different answers.

On circulation, the evidence is strong. The value settled over UPI rose from about half of nominal GDP in FY 2022-23 to about nine-tenths of it in FY 2025-26. Over the decade the number of payments rose thirteen times, while the average payment shrank by nearly a third. Read through Fisher's equation, that is a big rise in the number of transactions supported by a given stock of money. Read through the Cambridge version, it is a fall in the balance people need to hold against their spending. Both readings describe a rise in the transactions velocity of money. The qualification, which I state plainly in Data Set 4, is that part of the measured rise is cash payments moving into a system that counts them. Those payments were always real. They were simply invisible before.

On economic activity the evidence points one way but does not prove the case, and I do not claim more than that. The fact that 57 per cent of small shopkeepers report higher sales after taking up digital payment [15] is the most direct sign we have that the effect reaches output. The BIS finding that fast payment systems raise digital finance use most in lower-income countries [20], and the IMF finding of faster growth in districts where interoperability helped most [18], both point the same way. But none of this separates UPI's own effect from everything else that happened in the same decade. Smartphones, internet access, bank accounts and incomes all spread at the same time. A study based on published national figures cannot separate them, and it would be dishonest to pretend otherwise.

What I can say without any qualification is that UPI has changed how India pays. It did this by solving a problem that cards and wallets never could. It made the very small payment worth making. The average payment fell from ₹1,838 to ₹1,300 while volume rose more than ten times, and it did so even though inflation was pushing the other way. That is the sign of a system reaching payments it never served before. The tea, the vegetables, the auto fare: these are the payments UPI added, and they are why the national figures look the way they do.

So the case for greater efficiency is genuine. Money that does not have to be withdrawn, carried and deposited again is money free to be used. A shopkeeper paid at once can buy new stock sooner. A payment that leaves a record can later become the basis of a loan. These are real gains, and they fit everything I found in Chapter 4.

Two problems remain, and neither is small. The first is unequal access. Rural internet reaches less than half the urban rate. A payment system that needs a network cannot include more people than the network does, and the people left out are those who would gain most. The second is security. The rise in cyber incidents tracks the growth of the system, and the weak point is the user rather than the technology.

One last point is worth making. India built this system as public infrastructure and chose to make it free to use. The cost is paid from government funds rather than charged to shopkeepers. Almost every similar system abroad chose the opposite. The growth I have described is, in large part, a result of that decision. So the most important thing about UPI may not be its technology at all. It may be the idea that payment is a public service, which the technology was built to serve.

Glossary of Key Terms

Every technical term used in this report is explained below, in the sense in which I use it. The terms are in alphabetical order.

Term	What it means in this report
Average ticket size	The average value of one payment. You get it by dividing total value by total volume. I worked this out myself in Table 2. It is not published by any source.
Base year revision	A change in the reference year used to measure national income. India moved from a 2011-12 base to a 2022-23 base during my study period. So GDP figures from before and after the change do not match perfectly.
CAGR	Compound Annual Growth Rate. It is the steady yearly rate that would take a number from its starting value to its final value over the period.
Cambridge cash-balance approach	The formula $M = k \times P \times Y$. It treats the demand for money as the share of income people choose to hold as money. Velocity is one divided by k .
Currency in Circulation (CIC)	The total value of notes and coins held by the public and in bank tills. India's CIC compared with GDP is high by world standards, and the RBI wants to bring it down.
Demonetisation	The withdrawal of legal tender status from ₹500 and ₹1,000 notes on 9 November 2016. The notes were worth ₹15.4 trillion, which was 86.9 per cent of the value of all notes then in circulation.
Digital public infrastructure	Open digital systems that anyone can join and that serve the whole country. The Bank for International Settlements treats fast payment systems like UPI as the clearest example.
Equation of exchange	Fisher's equation $M \times V = P \times T$. It links the money stock and its velocity to the price level and the number of transactions. It is true by definition, not by assumption.
Fast payment system	A retail payment system that settles instantly or almost instantly, and runs all the time. UPI is the largest one in the world by number of payments.
Financial inclusion	Being able to use financial services that are useful and affordable. In India it is measured by the RBI's Financial Inclusion Index, which has three parts: access, usage and quality.
Formalisation	Economic activity moving from unrecorded to recorded. A digital payment leaves a record, and records build up into a credit history.

Term	What it means in this report
Income velocity	How many times a rupee is used to buy newly produced goods and services. You find it by dividing national income by the money stock. It is the safer measure of velocity, and it moves slowly.
Interoperability	Being able to pay a user of any other app or bank, not only one on the same service. The IMF says this is the design feature that made UPI spread.
k (cash-balance ratio)	The share of income people choose to keep as money. A fall in k means exactly the same thing as a rise in velocity.
Merchant Discount Rate (MDR)	The fee a shopkeeper pays on each digital payment. For UPI it is zero on small-merchant payments up to ₹2,000. The Government pays 0.15 per cent of the value instead.
Natural experiment	A real event that changes conditions sharply, without a researcher planning it. This lets us compare before and after. I treat demonetisation as one in Data Set 5.
Nominal GDP	The value of what the country produces, measured at current prices. It rises with both output and inflation. I use it as the bottom of the ratio in Table 4.
NPCI	The National Payments Corporation of India. It is the not-for-profit body that built and runs UPI, writes its rules, and settles money between member banks.
P2M and P2P	Person-to-merchant and person-to-person payments. P2M is now 63 per cent of UPI volume, and 86 per cent of P2M payments are below ₹500.
PIDF	The Payments Infrastructure Development Fund. It pays part of the cost of putting QR codes and card machines into tier-3 to tier-6 towns.
RBI Digital Payments Index	An RBI index that measures how far payments have gone digital. March 2018 is set at 100. It stood at 493.22 in March 2025.
Real-time settlement	A payment that is completed within seconds of being sent, instead of being processed in batches at fixed times.
Secondary data	Data collected and published by someone other than the researcher. My whole study uses secondary data, and I list this as its first limitation.
Structural break	A point where the relationship between two things changes, usually because of a shock or a new policy. COVID-19 is treated as one in the research in Section 3.6.
Transactions velocity	How many times a rupee is used in any payment at all. This includes transfers that do not buy anything new. Payment systems affect this measure directly, and it can move much

Term	What it means in this report
	faster than income velocity.
UPI	The Unified Payments Interface. It is a common set of rules, launched in 2016, that lets any bank account be paid from any app using a single short address.
Velocity of money	The average number of times one rupee is used to buy goods and services in a given period. It describes behaviour, not a physical quality of money.
Virtual Payment Address	A short name that stands in for a bank account number and branch code when making a UPI payment.

A note on one confusing abbreviation. DPI is used in two different ways in writing about payments. It can mean "digital public infrastructure" or the RBI's "Digital Payments Index". I have written both out in full everywhere they appear.

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